

Unit-1 Short answer questions

1. Define singular matrix
2. Define symmetric and skew symmetric matrices
3. Explain one hot encoding with example
4. List out the components of Biological neuron.
5. What are the components in a computational unit?
6. Difference between linearly separable and non-linearly separable datasets
7. What is the significance of the Perceptron Convergence Theorem?

Essay questions

- 1) Explain various types of matrices with examples
- 2) Explain the steps involved in PCA algorithm
- 3) Explain the structure of a biological neuron with a neat diagram.
- 4) What is a perceptron? Explain the steps involved in training a perceptron in deep learning
- 5) State and explain Perceptron Convergence Theorem, Explain why perceptron cannot solve XOR .

Unit -2 : Short answer questions

1. Define multi-layer perceptron
2. What are the three variants of Gradient Descent
3. List out any three activation functions used in hidden layer
4. What are various types of Loss functions and their suitable applications.
5. Define overfitting and underfitting
6. Give the list of types of Regularization methods.

Essay questions :

1. Explain about back propagation algorithm for training a neural network
2. What is a feed forward neural network? Explain gradient based learning in a feedword network
3. Explain the difficulties in training deep neural networks.
4. Explain various types of regularization and how does Regularization helps in reducing overfitting
5. Explain the steps involved in Greedy layerwise training of deep neural network.
6. Compute the output by applying sigmoid function, relu and tanh functions for the following input vector, weight vector and bias.

$$X = [0.5, 0.6, 0.1] \quad , \quad W = [0.2, 0.9, 0.3] \quad , \quad \text{bias } b = 0.5$$

Unit -3: Short answer questions

- 1) What is the main advantage of adaptive optimization methods?
- 2) What is Adam?
- 3) What is AdaGrad?
- 4) What is AdaDelta?
- 5) What is the purpose of dropout?
- 6) List out various Optimization methods

Essay questions :

1. What is the purpose of optimization in neural networks?
2. What is an optimization algorithm? What is the role of the learning rate in optimization?
3. Explain AdaGrad with its working.
4. Compare AdaGrad and AdaDelta
5. Differentiate Dropout and DropConnect.